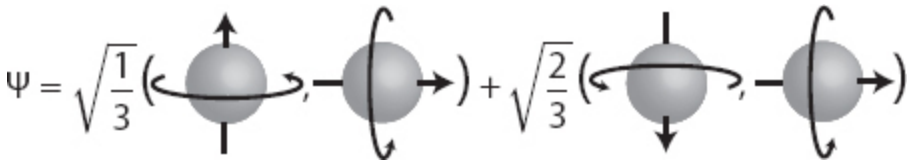
two choices; really, any reasonable rule should assign them equal probability.*

One nice thing about this more modest proposal is that no inconsistency arises with repeated measurements. Doing an extra measurement on one branch but not the other would leave us with branches that have unequal amplitudes again, so the rule doesn't seem to say anything at all.

But in fact it's way better than that. If we start with this simple equal-amplitudes-imply-equal-probabilities rule, and ask whether that is a special case of a more general rule that never leads to inconsistencies, we end up with a unique answer. And that answer is the Born rule: probability equals amplitude squared.

We can see this by returning to our unbalanced case, with one amplitude equal to the square root of 1/3 and the other equal to the square root of 2/3. This time we'll explicitly include a second horizontal spin-right qubit from the start. At first, this second qubit just goes along for the ride.



Insisting on equal probability for equal amplitudes doesn't tell us anything yet, since the amplitudes are not equal. But we can play the same game we did before, measuring the second spin along the vertical axis if the first spin is down. The wave function evolves into three components, and we can figure out what their amplitudes are by looking back at the decomposition of a spin-right state into vertical spins above. Multiplying the square root of 2/3 by the square root of 1/2 gives the square root of 1/3, so we get three branches, all with equal amplitudes.



Since the amplitudes are equal, we can now safely assign them equal probabilities. Since there are three of them, that's 1/3 each. And if we don't want the probability of one branch to suddenly change when something happens on another branch, that means we should have assigned probability 1/3 to the spin-up branch even before we did the second measurement. But